ECE 16 Final Project

Distraction Detector



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Agenda

1. **What is the Need?**
   - The use case
   - The problem's worth
   - Intended user(s)
2. **Intro to Our Solution**
   - A quick intro to our distraction detector
3. **How Does it Help?**
   - How it helps our intended users(s)
   - Addressing the user's need
4. **How We Did It?**
   - Our design choices
   - Our thought process
5. **Demo**
   - Show off our product



What is the need?

Use Case to Address

Distraction is a part of the human existence and is inevitable for most humans. Many times distraction can be a good thing especially after a full day of nonstop work. However, distraction could also mean disruption and it can even mean the difference between life and death. We want to address the need for something to detect distraction as well as the need to punish the distraction in order to mitigate negative outcomes.

Why this is Important

01 Many people have the habit of texting while driving

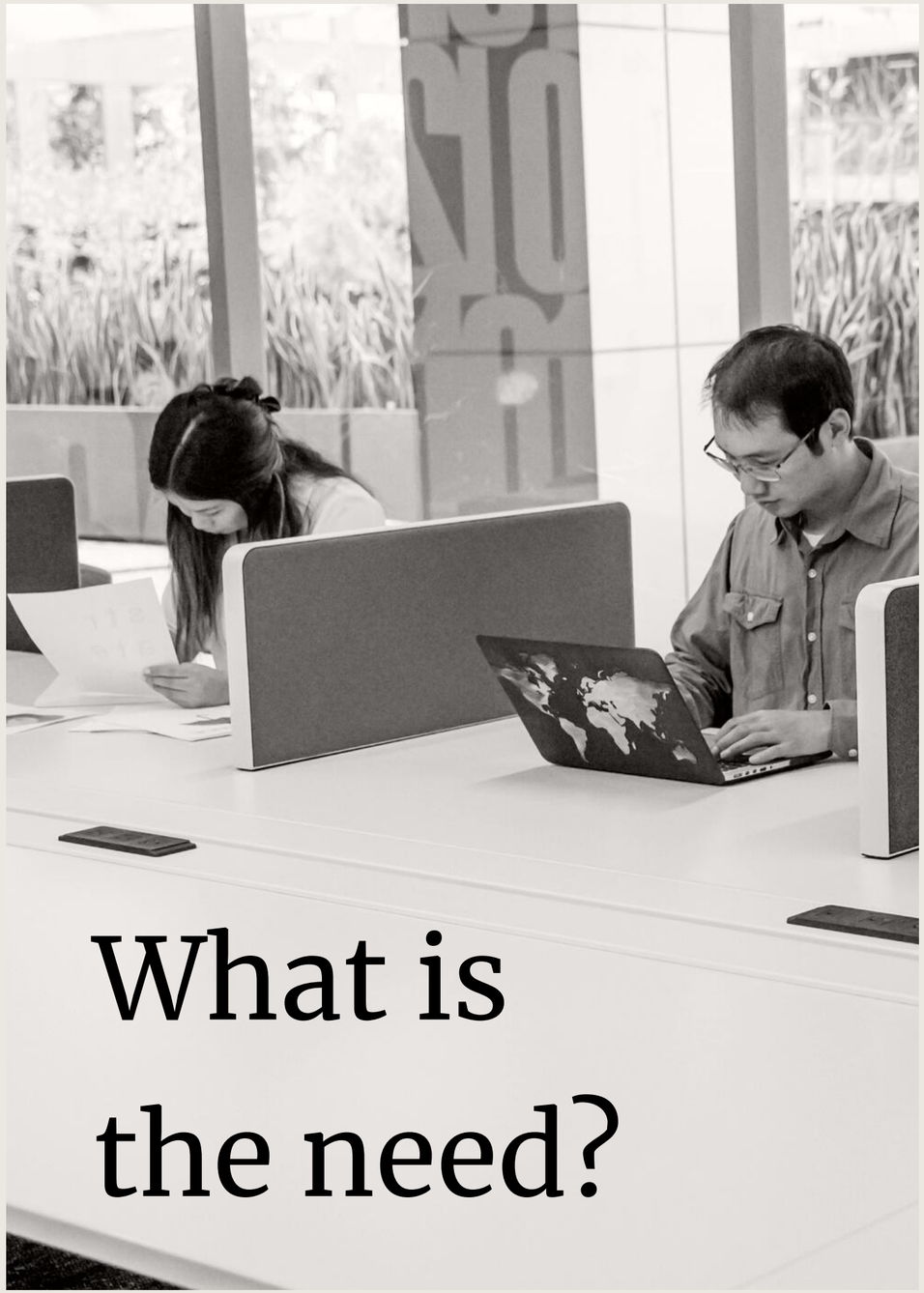
"According to the National Safety Council, mobile use while driving leads to 1.6 million accidents every year"

02 In the day and age of the pandemic, online exams have been more prevalent and online proctoring has been necessary to catch/eliminate cheating

"They were so desperate to connect that they were using — or in some courses being encouraged to create — group chats. Those chats then became a place where they may talk about homework or talk about exam questions." - npr.org

03 As a student, it is incredibly easy to lose time and sleep due to distraction

“Each time you switch your focus, you compromise your effectiveness at work. Even the smallest distractions reduce your productivity capacity, says Business News Daily”



What is the need?

Intended Users

01 Car Manufacturers/Drivers



02 Professors/TAs

UCSD

03 Students/Workers

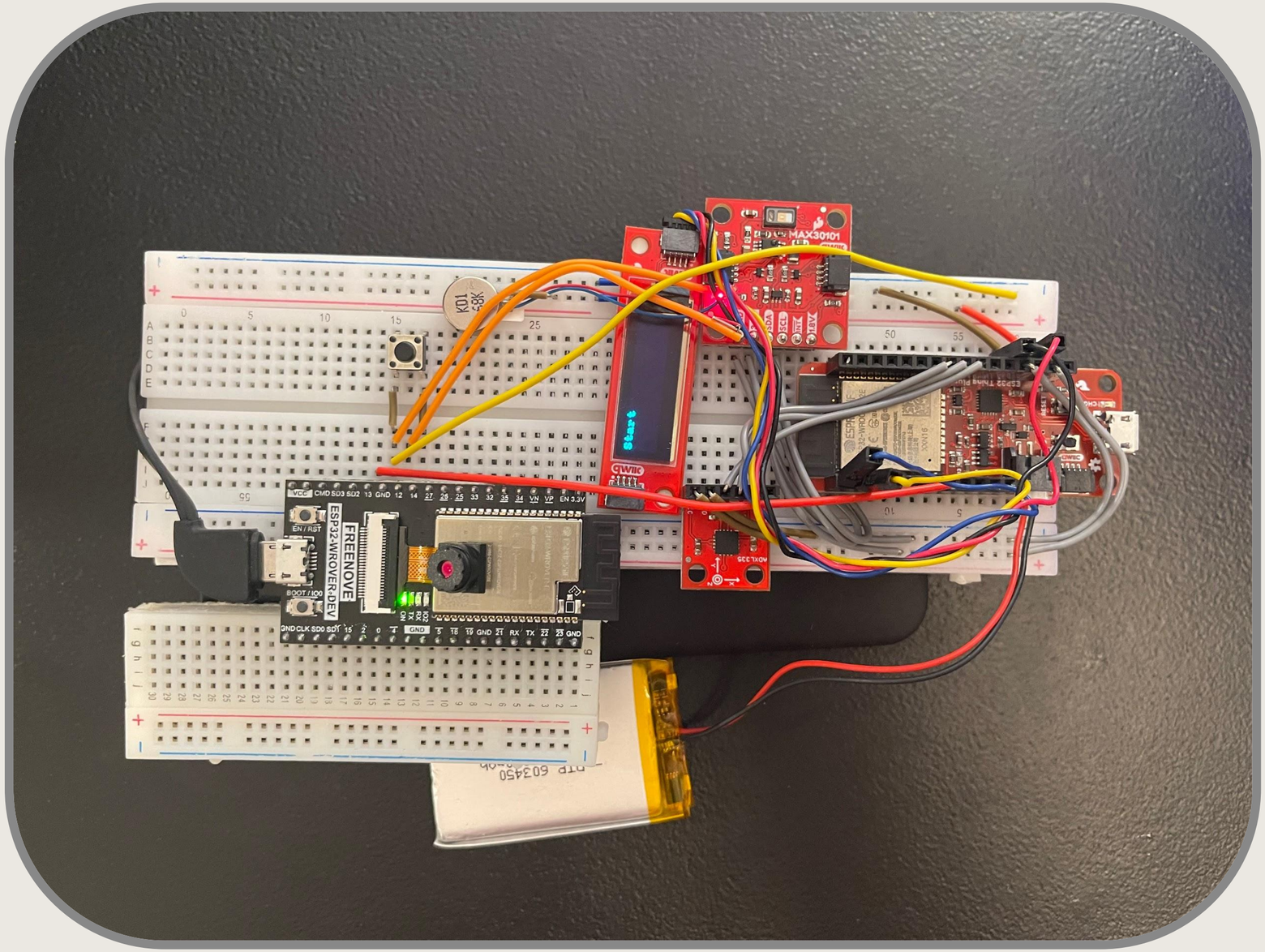


Our Solution

Phone Detector

The phone was the common denominator between all the negative outcomes that come from distraction

It is a portable solution that can be used in many environments



How it Helps Car Manufacturers



Helps mitigate car accidents as there will be direct feedback when a driver either subconsciously or consciously pull out their phone



Car manufacturers can benefit from the trust that buyers will gain when they realize our product is there to provide for an extra safety feature.

How it helps Professors/TA’s

Online Proctoring



Watch over student’s workspace

Prevent Unauthorized Tools



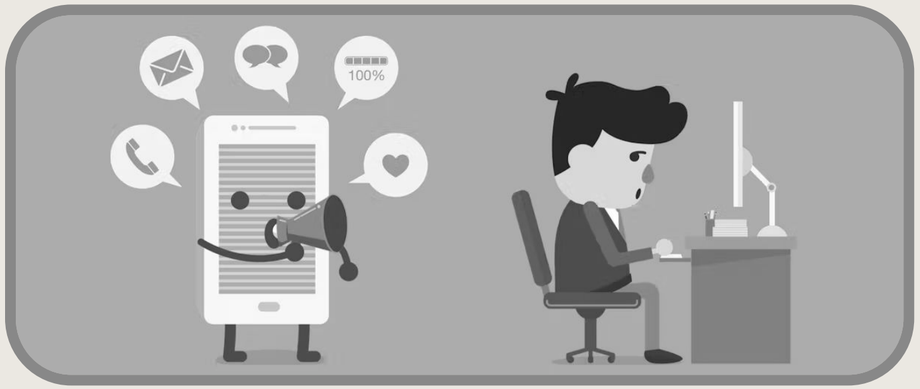
Ensure that they don’t pull out tools that are not allowed i.e. Phone

Mitigate Collaboration

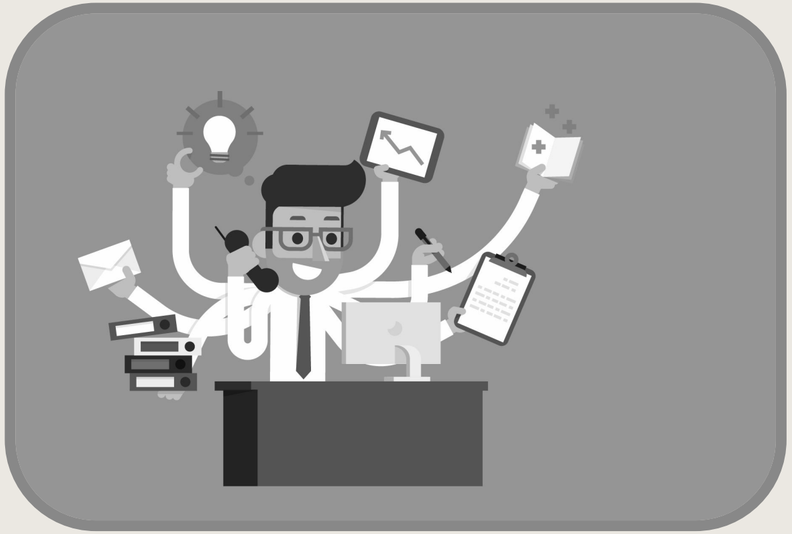


The phone is one way students collaborate online to cheat off of each other

How it Helps Students/Workers

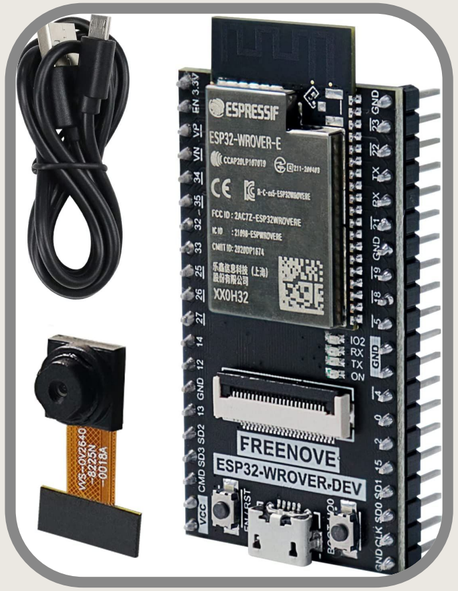


The device can help students and workers focus and realize when they are distracted as distraction can come automatically or subconsciously

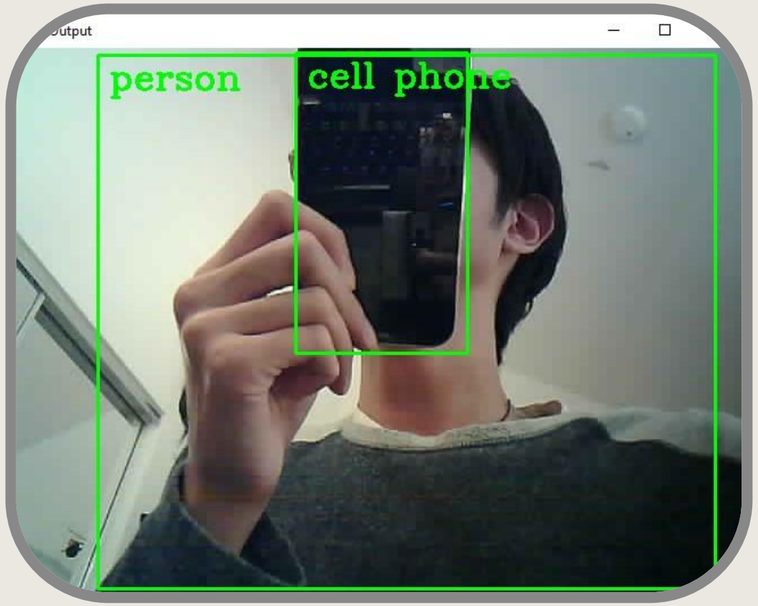


As a result of immediate feedback whenever a student or worker is distracted, their productivity levels can be boosted

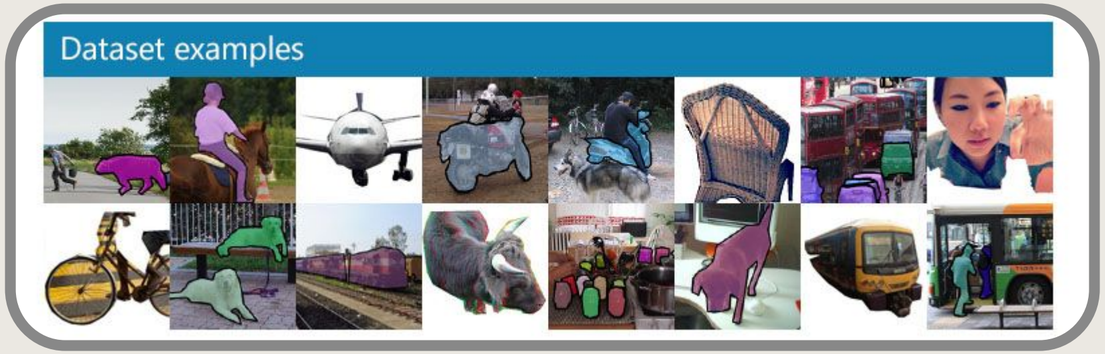
How Do We do it?



ESP32cAM



Phone Detector



COCO Dataset

- Setup the ESP32CAM as an access point that will transmit video feed
- Python reads in this video feed and uses OpenCV in order to perform object detection
- The detection model was pre-trained with the famous COCO dataset
- Non-Maximum suppression is used to ensure accuracy and that there are no duplicates
- Python will send a “detected” or “undetected” message to the original ESP32 board whenever a phone is detected or not
- The main board sends a signal to the ESP32CAM board to signify when to buzz the buzzer
- A button is implemented to turn the feature on or off



DEMO